

GEORGE LARSON

Email: george.g.larson@gmail.com · Phone: 412-945-0101 · Location: Pittsburgh, PA

Director of Technology with 25+ years leading software teams in manufacturing, logistics, and ecommerce operations. Built and scaled platforms handling production-floor integrations, warehouse management, ERP systems, and fulfillment pipelines across print, ecommerce, and medical billing industries. Hands-on with C#, .NET Core, SQL, and RESTful APIs throughout my career — not just directing, but writing the code that connects production systems to business outcomes. Led teams from 10 to 50+ across internal, offshore, and vendor resources.

CORE STRENGTHS

Technology Leadership

- Strategic roadmaps that connect executive goals to team-level delivery.
- Calm stewardship for blended internal, offshore, and vendor teams (Sony, USC, FM Expressions).
- Agile delivery, ticketing, and analytics to align product, operations, and finance on throughput, defect, and margin targets.

Software Development

- Full stack delivery in C#, .NET Core, SQL Server, MySQL, PostgreSQL, RESTful APIs, React, TypeScript, PHP, Python, and Go.
- Code review, automated testing, and performance tuning for legacy and greenfield systems.
- ERP and WMS integration: wrote software that directly interfaces with warehouse management, inventory, and order fulfillment systems.

Operations and Logistics

- Production-floor system integration: print-head control software, PLC programming, conveyor and packaging automation.
- Fulfillment vendor stack integration (Pace/EFI, Printro, VivX, Fanatics, Alphabroder, SignTronics) with ecommerce platforms (Shopify, Magento).
- Parts ordering, inventory management, and supply-chain tooling (AS/400, VT-52 terminal emulation, logistics operations at Sony).
- OCR digitization, ETL pipelines, and knowledge-base design for operational data.

DevOps and Infrastructure

- CI/CD design, infrastructure as code, container orchestration, and repeatable releases.
- Observability, incident response, high-availability design (replication, backups, CDN), and SLO tracking.
- Database performance tuning: improved query performance by 93%, cut slow-query volume by 96%.

Security and Compliance

- Network hardening, IAM, zero-trust rollouts, and automated vulnerability assessment loops.
- HIPAA/HITECH compliance (medical billing), CCPA/GDPR compliance work, regulated-data guardrails.

WORK EXPERIENCE

Director of Technology, FM Expressions (2018 to 2026)

- Led technology strategy for a mid-size manufacturing company serving ecommerce channels, overseeing all software development, infrastructure, and vendor integrations.
- Built and operated the print-head control software in C#/.NET Core, including network traffic capture and protocol analysis to secure industrial print systems and debug production-line issues.
- Developed Printro integrations in C#/.NET Core connecting the production floor to the order pipeline, enabling real-time fulfillment across multiple vendor channels.
- Built and operated integrations across the full print and fulfillment vendor stack (Pace/EFI, Printro, VivX, Fanatics, Alphabroder, SignTronics) plus ecommerce platforms (Shopify, Magento) and marketing tooling (HubSpot, Power BI), so the production floor, order pipeline, and analytics layer share a coherent data model.
- Oversaw ERP and WMS solutions: wrote software that directly interfaced with warehouse management, inventory, and order fulfillment systems, driving the production floor's data model from order intake through shipping.

- Improved database performance by 93% while driving critical systems to 99% uptime; cut MySQL slow-query volume by 96% to accelerate incident response.
- Captured Meteor print-head PCAPs and tuned security assessments to secure industrial protocols while migrating remote access from VPN to zero-trust policies.
- Built financial-modeling tools using AI/ML to analyze company performance and forecast scenarios; presented findings directly to executive leadership.
- Managed all DevOps, CI/CD, infrastructure, and security operations: GitLab (36 projects, 739 CI pipelines), high-availability platforms (dual MySQL replicas, VMware ESXi, Datto backups), and security hardening (36 public assets scanned, zero critical/high issues in backlog).
- Implemented Agile delivery, ticketing, and analytics to align product, operations, and finance on throughput, defect, and margin targets.

Senior Software Engineer and Manager, Universal SmartComp (2009 to 2016)

- Helped scale the company from a 50-person startup to a national leader in medical billing (Inc 5000, ranked #34 in Health and #503 overall).
- Designed the EDI framework, ACH transmission, ERP integration, payer portal, Gold Standard Review, and recovery analytics layers of the platform.
- Led internal and offshore teams across development and Linux infrastructure.
- Acted as security response lead, HIPAA/HITECH SME, and Linux SME for the medical billing platform.
- Deployed VMware ESXi covering web, database, DHCP, DNS, and ERP servers across the platform.
- Built OCR bill entry validation that reduced errors and sped turnaround.

Software Engineer and Team Manager, Sony Electronics (2004 to 2009)

- Directed software development, logistics, and technical operations for the Pittsburgh Customer Satisfaction Center — joined a service operation that was effectively bare shelves at start and built it into a functioning logistics hub.
- Managed all inbound and outbound logistics: parts ordering, inventory management, repair tracking, and shipping across the US market.
- Created an AS/400 parts ordering interface that saved millions in operational spend by streamlining the supply chain from order to delivery.
- Developed a VT-52 terminal emulation system used for inventory management across the warehouse.
- Designed and ran the TiVo firmware upgrade station for the United States market — a logistics operation at national scale.
- Built hardware testing suites for CD drives, set-top boxes, PCBs, televisions, cameras, and MP3 players.
- Front-end software work delivered roughly 30% productivity gain and 12% expense reduction.
- Managed more than 50 combined technical and non-technical staff, including roughly 30 contracted personnel.

Maintenance and Automation Technician, T. Marzetti Company (2000 to 2004)

- Led second shift maintenance for robotics, conveyors, and packaging machinery.
- Programmed PLCs and Adept V++ robotics in-house, tuned sensors, actuators, and industrial controls on the production line.
- Kept robotics development internal to limit vendor cost.

Earlier Career Highlights (1990s to 2000s)

- Contract developer for Daifuku USA working on material handling support systems.
- Software developer and webmaster for non-profit organizations.
- Engineer for The Sounding Board, a psychotherapeutic BBS in Columbus, Ohio; built custom C++ modules for Major BBS.
- Paid programming and database work 1994 to 2000 via Pacific Judgment Recovery, AGL Personnel, and Remedy Staffing CNC contract.

TECHNICAL SKILLS

Languages: C#, .NET Core, SQL, TypeScript, React, PHP, Python, Go, Rust, Bash, C++

Databases: SQL Server, MySQL, PostgreSQL

Web and APIs: RESTful APIs, ASP.NET, Node.js, GraphQL

Infrastructure: Linux, Windows Server, IIS, Nginx, Caddy, Kubernetes, VMware ESXi, Datto backups

Cloud: AWS, Azure, Google Cloud

DevOps: GitLab CI/CD, GitHub Actions, Docker, infrastructure as code

Security: Authentik, HashiCorp Vault, CrowdSec, HostedScan, OWASP ZAP, Nikto, Metasploit, SSL/TLS, iptables, UFW, fail2ban, zero-trust network design, HIPAA/HITECH, CCPA/GDPR

Ecommerce and Operations: Shopify, Magento, ERP integration, WMS integration, fulfillment vendor integrations, Power BI, HubSpot

Manufacturing: PLC programming, industrial controls, print-head control software, conveyor and packaging automation, production-floor system integration

Monitoring: Uptime Kuma, HostedScan, Red Sift OnDMARC, Dashy

EDUCATION AND PROFESSIONAL DEVELOPMENT

- Self-taught from C++ books as a teenager; built BBS modules in production before attending college.
- Continuous learner across hundreds of online courses, lectures, and certifications (see LinkedIn for details).
- Additional coursework spanning systems engineering, network security, database internals, and cloud architecture through platforms including Coursera, MIT OpenCourseWare, and vendor certification programs.